```
> restart; #WaveElect;
```

```
> eq1 := 2·m/h^2 (E + e0^2/r)·f(r,t) + R/r diff((f(r,t)), r$2) − 1/(a^2·c^2) diff(f(r,t), t$2) = 0;
```

$$eq1 := \frac{2\,m\left(E+\frac{e0^2}{r}\right)f(r,t)}{h^2}+\frac{R\left(\frac{\partial^2}{\partial r^2}\,f(r,t)\right)}{r}-\frac{\frac{\partial^2}{\partial t^2}\,f(r,t)}{a^2\,c^2}=0 \tag{1}$$

```
> iv1 := f(R,t) = 0, D[1](f)(0,0) = R, f(rb,0) = 0, D[2](g)(rb,0) = a·c;
```

$$iv1 := f(R,t)=0,\ D_1(f)(0,0)=R,\ f(rb,0)=0,\ D_2(g)(rb,0)=a\,c \tag{2}$$

```
> change_of_vars := { r = rb·rho, f(r,t) = g(rho,t)·(R − rb·rho)/R };
```

$$change_of_vars := \left\{r=rb\,\rho,\ f(r,t)=\frac{g(\rho,t)\,(-rb\,\rho+R)}{R}\right\} \tag{3}$$

```
> tmp := PDEtools:-dchange(change_of_vars, [eq1, iv1], [g(rho,t), rho]);
```

$$tmp := \left[\frac{2\,m\left(E+\frac{e0^2}{rb\,\rho}\right)g(\rho,t)\,(-rb\,\rho+R)}{h^2\,R}+\frac{R\left(\frac{\left(\frac{\partial^2}{\partial\rho^2}\,g(\rho,t)\right)(-rb\,\rho+R)}{R}-\frac{2\left(\frac{\partial}{\partial\rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho}-\frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+R)}{a^2\,c^2\,R}\right.$$
$$\left.=0,\ 0=0,\ \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{R}}{rb}=R,\ \frac{g(1,0)\,(R-rb)}{R}=0,\ D_2(g)(rb,0)=a\,c\right] \tag{4}$$

```
> eq2 := tmp[1];
```

$$eq2 := \frac{2\,m\left(E+\frac{e0^2}{rb\,\rho}\right)g(\rho,t)\,(-rb\,\rho+R)}{h^2\,R}+\frac{R\left(\frac{\left(\frac{\partial^2}{\partial\rho^2}\,g(\rho,t)\right)(-rb\,\rho+R)}{R}-\frac{2\left(\frac{\partial}{\partial\rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho}-\frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+R)}{a^2\,c^2\,R}=0 \tag{5}$$

```
> eq3 := tmp[3];
```

$$eq3 := \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{R}}{rb}=R \tag{6}$$

```
> eq4 := tmp[4];
```

$$eq4 := \frac{g(1,0)\,(R-rb)}{R}=0 \tag{7}$$

```
> eq5 := tmp[5];
```

$$eq5 := D_2(g)(rb,0)=a\,c \tag{8}$$

```
> iv2 := D_1(g)(0,0) − g(0,0)·rb/R = rb·R, g(1,0)·(R − rb)/R = 0, D_2(g)(rb,0);
```

$$iv2 := D_1(g)(0,0)-\frac{g(0,0)\,rb}{R}=rb\,R,\ \frac{g(1,0)\,(R-rb)}{R}=0,\ D_2(g)(rb,0) \tag{9}$$

```
> sys := [eq2, iv2]: Sol := pdsolve(sys, HINT = `*`);
```

$$Sol := g(\rho, t) = -\left(rb\,R^2 \left(-_C3^2 \left(e^{\sqrt{-c_2}\,t} + e^{-\sqrt{-c_2}\,t} \right) \left(\mathrm{AiryBi}\left(-\frac{rb^2 \left(2\,E\,a^2 c^2 m\,rb + 2\,a^2 c^2 e0^2 m - h^2 rb\,_c_2 \right)}{\left(\frac{rb^3 \left(2\,c^2 E m a^2 - _c_2 h^2 \right)}{R\,a^2 c^2 h^2} \right)^{2/3} c^2 a^2 R h^2} \right) _C3^2 \right.\right.\right.$$

$$\left.\left.\left. + \frac{\mathrm{AiryBi}\left(-\frac{\left(\frac{rb^3 E m}{R h^2} \right)^{1/3} 2^{1/3} \left(E\,rb + e0^2 \right)}{E\,rb} \right) R\,rb}{2} \right) \mathrm{AiryAi}\left(-\frac{\left(\frac{rb^3 \left(2\,c^2 E m a^2 - _c_2 h^2 \right)}{R\,a^2 c^2 h^2} \right)^{1/3} \left(2\,E\,a^2 c^2 m\,rb\,\rho + 2\,a^2 c^2 e0^2 m - rb\,\rho\,_c_2 h^2 \right)}{rb \left(2\,c^2 E m a^2 - _c_2 h^2 \right)} \right) \right.$$

$$\left. + _C3^2 \left(e^{\sqrt{-c_2}\,t} + e^{-\sqrt{-c_2}\,t} \right) \left(\mathrm{AiryAi}\left(-\frac{rb^2 \left(2\,E\,a^2 c^2 m\,rb + 2\,a^2 c^2 e0^2 m - h^2 rb\,_c_2 \right)}{\left(\frac{rb^3 \left(2\,c^2 E m a^2 - _c_2 h^2 \right)}{R\,a^2 c^2 h^2} \right)^{2/3} c^2 a^2 R h^2} \right) _C3^2 \right.\right.$$

$$\left.\left. + \frac{R\,rb\,\mathrm{AiryAi}\left(-\frac{\left(\frac{rb^3 E m}{R h^2} \right)^{1/3} 2^{1/3} \left(E\,rb + e0^2 \right)}{E\,rb} \right)}{2} \right) \mathrm{AiryBi}\left(-\frac{\left(\frac{rb^3 \left(2\,c^2 E m a^2 - _c_2 h^2 \right)}{R\,a^2 c^2 h^2} \right)^{1/3} \left(2\,E\,a^2 c^2 m\,rb\,\rho + 2\,a^2 c^2 e0^2 m - rb\,\rho\,_c_2 h^2 \right)}{rb \left(2\,c^2 E m a^2 - _c_2 h^2 \right)} \right) - rb\,R \left(\right.\right.$$

$$\left.\left. -\mathrm{AiryBi}\left(-\frac{2^{1/3} \left(\frac{rb^3 E m}{R h^2} \right)^{1/3} \left(E\,rb\,\rho + e0^2 \right)}{E\,rb} \right) \mathrm{AiryAi}\left(\right.\right.\right. \right) \tag{10}$$

$$-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)\,_C3^2+\mathrm{AiryAi}\Bigg($$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\Bigg)\,\mathrm{AiryBi}\Bigg($$

$$-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)\,_C3^2-\frac{1}{2}\Bigg(R\,rb\,\Bigg(\mathrm{AiryBi}\Bigg($$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\Bigg)\,\mathrm{AiryAi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)-\mathrm{AiryAi}\Bigg($$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\Bigg)\,\mathrm{AiryBi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)\Bigg)\Bigg)\Bigg)\Bigg)$$

$$\Bigg/\Bigg(2\Bigg(\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,\mathrm{AiryBi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)R\,rb\,_C3^2}{2}$$

$$-\mathrm{AiryBi}\Bigg(1,-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\Bigg)\Bigg(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\Bigg)^{1/3}_C3^4\Bigg)$$

$$\mathrm{AiryAi}\left(-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)$$

$$+_C3^2\left(\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,R\,rb\,\mathrm{AiryAi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)}{2}+\mathrm{AiryAi}\left(1,\right.\right.$$

$$\left.\left.-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}_C3^2\right)\mathrm{AiryBi}\left(\right.$$

$$\left.-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)-\frac{1}{2}\left(\left(\left(-\mathrm{AiryBi}\left(\right.\right.\right.\right.$$

$$\left.\left.\left.\left.-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\right)\mathrm{AiryAi}\left(1,\right.\right.\right.\right.$$

$$\left.\left.\left.\left.-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}_C3^2+\mathrm{AiryBi}\left(1,\right.\right.\right.\right.$$

$$-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\Bigg)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\mathrm{AiryAi}\Bigg($$

$$-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)_C3^2-\frac{1}{2}\Bigg(\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}R\,rb\Bigg(\mathrm{AiryAi}\Bigg(1,$$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\Bigg)\mathrm{AiryBi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)-\mathrm{AiryBi}\Bigg(1,$$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\Bigg)\mathrm{AiryAi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)\Bigg)\Bigg)\Bigg)rb\,R\Bigg)\Bigg)\left(rb\,\rho\right.$$

$$\left.-R\right)\Bigg)$$

> $G := rhs(Sol);$

$$G := -\Bigg(rb\,R^2\Bigg(-_C3^2\left(e^{\sqrt{-c_2}\,t}+e^{-\sqrt{-c_2}\,t}\right)\Bigg(\mathrm{AiryBi}\Bigg($$

$$-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)_C3^2$$

$$+\frac{\mathrm{AiryBi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\right)R\,rb}{2}\Bigg)\mathrm{AiryAi}\Bigg($$

$$-\frac{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(2\,E\,a^2\,c^2\,m\,rb\,\rho+2\,a^2\,c^2\,e0^2\,m-rb\,\rho\,_c_2\,h^2\right)}{rb\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}\Bigg)$$

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$$+_C3^2\left(e^{\sqrt{-\frac{c}{2}}\,t}+e^{-\sqrt{-\frac{c}{2}}\,t}\right)\left(\mathrm{AiryAi}\left(-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)_C3^2\right.$$

$$\left.+\frac{R\,rb\,\mathrm{AiryAi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\right)}{2}\right)\mathrm{AiryBi}\left(\right.$$

$$\left.-\frac{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(2\,E\,a^2\,c^2\,m\,rb\,\rho+2\,a^2\,c^2\,e0^2\,m-rb\,\rho\,_c_2\,h^2\right)}{rb\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}\right)-rb\,R\left(\right.$$

$$-\mathrm{AiryBi}\left(-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\right)\mathrm{AiryAi}\left(\right.$$

$$\left.-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)_C3^2+\mathrm{AiryAi}\left(\right.$$

$$\left.-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\right)\mathrm{AiryBi}\left(\right.$$

$$\left.-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)_C3^2-\frac{1}{2}\left(R\,rb\left(\mathrm{AiryBi}\left(\right.\right.\right.$$

$$\left.\left.\left.-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\right)\mathrm{AiryAi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\right)-\mathrm{AiryAi}\left(\right.\right.\right.$$

$$\left.\left.\left.-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\right)\mathrm{AiryBi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\right)\right)\right)\right)\right)$$

$$\Bigg/\left(2\left(\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,\mathrm{AiryBi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)R\,rb\,_C3^2}{2}\right.\right.\right.$$

$$\left.\left.\left.-\mathrm{AiryBi}\left(1,-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}_C3^4\right)\right.\right.$$

$$\mathrm{AiryAi}\left(-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)$$

$$\left.\left.+_C3^2\left(\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,R\,rb\,\mathrm{AiryAi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)}{2}+\mathrm{AiryAi}\left(1,\right.\right.\right.\right.$$

$$\left.\left.\left.\left.-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}_C3^2\right)\mathrm{AiryBi}\left(\right.\right.\right.\right.$$

$$-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-h^2\,rb\,_c_2\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)-\frac{1}{2}\Bigg(\Bigg(\Bigg(-\mathrm{AiryBi}\Bigg($$

$$-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)\mathrm{AiryAi}\Bigg(1,$$

$$-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\Bigg)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}_C3^2+\mathrm{AiryBi}\Bigg(1,$$

$$-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\Bigg)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-_c_2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\mathrm{AiryAi}\Bigg($$

$$-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)_C3^2-\frac{1}{2}\Bigg(\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,R\,rb\,\Bigg(\mathrm{AiryAi}\Bigg(1,$$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\Bigg)\mathrm{AiryBi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)-\mathrm{AiryBi}\Bigg(1,$$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\Bigg)\mathrm{AiryAi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)\Bigg)\Bigg)\Bigg)rb\,R\Bigg)\Bigg)\left(rb\,\rho\right.$$

$$\left.-R\right)\Bigg)$$

> $H := subs\left(t=0,\ _C3=1,\ _c_2=a^2\cdot c^2,\ G\right);$

$$H := -\Bigg(rb\,R^2\Bigg(-2\,\mathrm{e}^0\Bigg(\mathrm{AiryBi}\Bigg(-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)\Bigg)\Bigg)\Bigg)\qquad\textbf{(12)}$$

$$+\frac{\mathrm{AiryBi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\,2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)R\,rb}{2}\Bigg)\,\mathrm{AiryAi}\Bigg($$

$$-\frac{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{1/3}(2\,E\,a^2\,c^2\,m\,rb\,\rho-a^2\,c^2\,h^2\,rb\,\rho+2\,a^2\,c^2\,e0^2\,m)}{rb\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}\Bigg)$$

$$+2\,\mathrm{e}^0\Bigg(\mathrm{AiryAi}\left(-\frac{rb^2\,(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb)}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)$$

$$+\frac{R\,rb\,\mathrm{AiryAi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\,2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)}{2}\Bigg)\,\mathrm{AiryBi}\Bigg($$

$$-\frac{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{1/3}(2\,E\,a^2\,c^2\,m\,rb\,\rho-a^2\,c^2\,h^2\,rb\,\rho+2\,a^2\,c^2\,e0^2\,m)}{rb\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}\Bigg)-rb\,R\Bigg($$

$$-\mathrm{AiryBi}\left(-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}(E\,rb\,\rho+e0^2)}{E\,rb}\right)\mathrm{AiryAi}\Bigg($$

$$-\frac{rb^2\,(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb)}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)+\mathrm{AiryAi}\Bigg($$

$$-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}(E\,rb\,\rho+e0^2)}{E\,rb}\Bigg)\,\mathrm{AiryBi}\Bigg($$

$$-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)-\frac{1}{2}\Bigg(R\,rb\Bigg(\mathrm{AiryBi}\Bigg(-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\Bigg)\mathrm{AiryAi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)-\mathrm{AiryAi}\Bigg(-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}\left(E\,rb\,\rho+e0^2\right)}{E\,rb}\Bigg)\mathrm{AiryBi}\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\left(E\,rb+e0^2\right)}{E\,rb}\Bigg)\Bigg)\Bigg)\Bigg)\Bigg)\Bigg/\Bigg(2\Bigg(\Bigg(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,\mathrm{AiryBi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)R\,rb}{2}$$

$$-\mathrm{AiryBi}\Bigg(1,-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\Bigg)\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\Bigg)$$

$$\mathrm{AiryAi}\Bigg(-\frac{rb^2\left(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb\right)}{\left(\frac{rb^3\left(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2\right)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\Bigg)$$

$$+\Bigg(\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,R\,rb\,\mathrm{AiryAi}\left(1,-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)}{2}+\mathrm{AiryAi}\Bigg(1,$$

$$-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\text{AiryBi}\left(-\frac{rb^2\,(2\,E\,a^2\,c^2\,m\,rb+2\,a^2\,c^2\,e0^2\,m-a^2\,c^2\,h^2\,rb)}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}c^2\,a^2\,R\,h^2}\right)-\frac{1}{2}\left(\left(-\text{AiryBi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)\text{AiryAi}\left(1,\right.\right.\right.$$

$$\left.-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{1/3}+\text{AiryBi}\left(1,\right.$$

$$\left.-\frac{2\,e0^2\,m\,rb^2}{\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{2/3}R\,h^2}\right)\left(\frac{rb^3\,(2\,c^2\,E\,m\,a^2-a^2\,c^2\,h^2)}{R\,a^2\,c^2\,h^2}\right)^{1/3}\text{AiryAi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)-\frac{1}{2}\left(\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,R\,rb\left(\text{AiryAi}\left(1,\right.\right.\right.$$

$$\left.-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)\text{AiryBi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)-\text{AiryBi}\left(1,\right.$$

$$\left.\left.\left.-\frac{2^{1/3}\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}e0^2}{E\,rb}\right)\text{AiryAi}\left(-\frac{\left(\frac{rb^3\,E\,m}{R\,h^2}\right)^{1/3}2^{1/3}\,(E\,rb+e0^2)}{E\,rb}\right)\right)\right)\right)\right)\,rb\,R\Bigg)\Bigg)\,(rb\,\rho$$

$$-R)\Bigg)$$

> $m := 9.109\cdot 10^{-31}$; $h := \frac{6.6256}{2\cdot \text{Pi}}\cdot 10^{-34}$; $e0 := 1.602\cdot 10^{-19}$; $c := 299792458$; $rb := 5.292410932\,10^{-11}$;

$$R := 3.566 \cdot 10^{22}; e := 2.71828; a := \frac{1}{137}; E := 2.179 \cdot 10^{-18};$$

$$m := 9.109000000\ 10^{-31}$$
$$h := 1.054496991\ 10^{-34}$$
$$e0 := 1.602000000\ 10^{-19}$$
$$c := 299792458$$
$$rb := 5.292410932\ 10^{-11}$$
$$R := 3.566000000\ 10^{22}$$
$$e := 2.71828$$
$$a := \frac{1}{137}$$
$$E := 2.179000000\ 10^{-18} \quad \textbf{(13)}$$

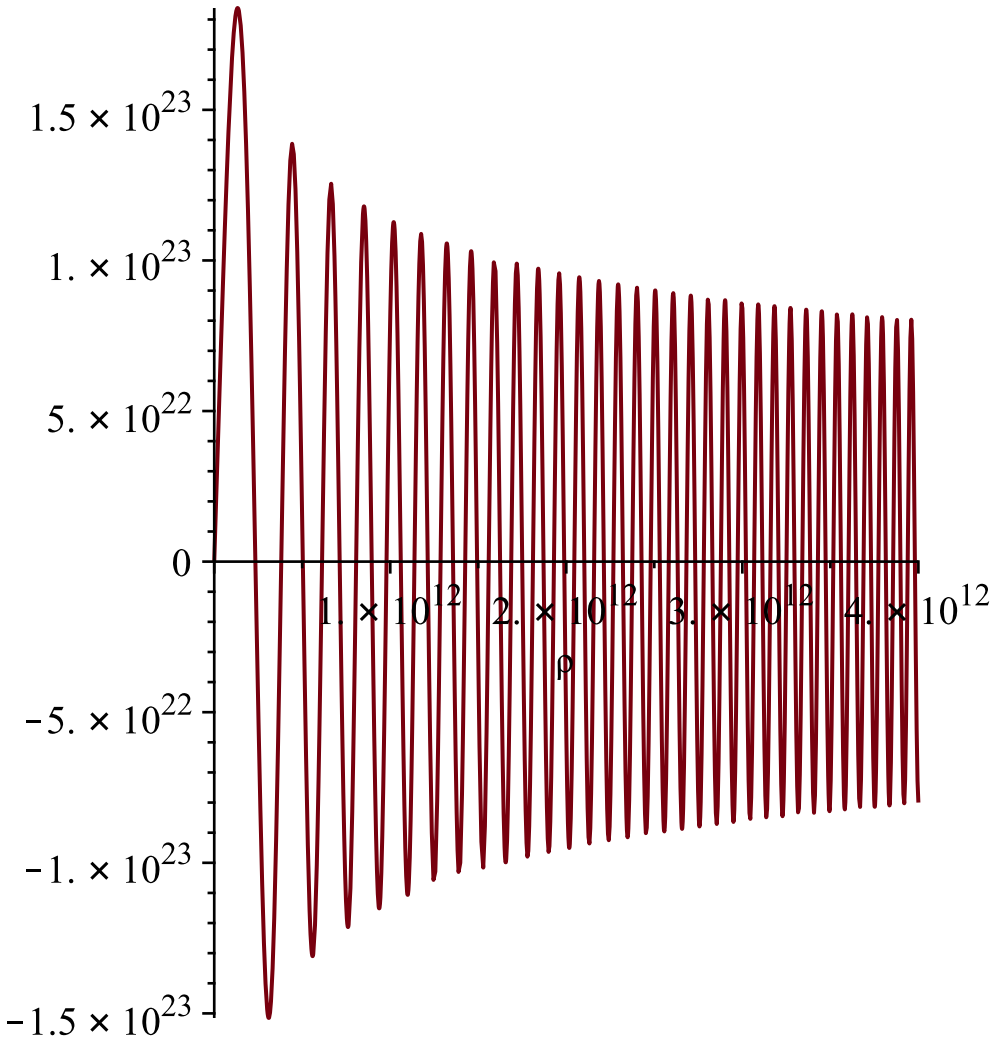
> H;

$$-(3.365009076\ 10^{34}\ (-2\ (0.6149266274 + 9.436368690\ 10^{11}\ \mathrm{AiryBi}(-9.053271108\ 10^{-12}\ 2^{1/3}))\ \mathrm{AiryAi}(-1.140640683\ 10^{-11}\ \rho - 2.538417733\ 10^{-21}) + 2\ (0.3550280539 + 9.436368690\ 10^{11}\ \mathrm{AiryAi}(-9.053271108\ 10^{-12}\ 2^{1/3}))\ \mathrm{AiryBi}(-1.140640683\ 10^{-11}\ \rho - 2.538417733\ 10^{-21}) + 6.700351224\ 10^{11}\ \mathrm{AiryBi}(-7.850453359\ 10^{16}\ 2^{1/3}\ (1.153216342\ 10^{-28}\ \rho + 2.566404000\ 10^{-38})) - 1.160534875\ 10^{12}\ \mathrm{AiryAi}(-7.850453359\ 10^{16}\ 2^{1/3}\ (1.153216342\ 10^{-28}\ \rho + 2.566404000\ 10^{-38})) + 1.780901081\ 10^{24}\ \mathrm{AiryBi}(-7.850453359\ 10^{16}\ 2^{1/3}\ (1.153216342\ 10^{-28}\ \rho + 2.566404000\ 10^{-38}))\ \mathrm{AiryAi}(-9.053271108\ 10^{-12}\ 2^{1/3}) - 1.780901081\ 10^{24}\ \mathrm{AiryAi}(-7.850453359\ 10^{16}\ 2^{1/3}\ (1.153216342\ 10^{-28}\ \rho + 2.566404000\ 10^{-38}))\ \mathrm{AiryBi}(-9.053271108\ 10^{-12}\ 2^{1/3}))) / ((-3.033004806\ 2^{1/3}\ \mathrm{AiryBi}(1, -2.014743490\ 10^{-21}\ 2^{1/3}) - 3.630772064\ 10^{-12} + 5.253318424\ 2^{1/3}\ \mathrm{AiryAi}(1, -2.014743490\ 10^{-21}\ 2^{1/3}) - 2.785804216\ \mathrm{AiryBi}(-9.053271108\ 10^{-12}\ 2^{1/3}) - 4.825154441\ \mathrm{AiryAi}(-9.053271108\ 10^{-12}\ 2^{1/3}) + 8.061490150\ 10^{12}\ 2^{1/3}\ (\mathrm{AiryAi}(1, -2.014743490\ 10^{-21}\ 2^{1/3})\ \mathrm{AiryBi}(-9.053271108\ 10^{-12}\ 2^{1/3}) - \mathrm{AiryBi}(1, -2.014743490\ 10^{-21}\ 2^{1/3})\ \mathrm{AiryAi}(-9.053271108\ 10^{-12}\ 2^{1/3})))\ (5.292410932\ 10^{-11}\ \rho - 3.566000000\ 10^{22}))) \quad \textbf{(14)}$$

> $solve(H = 0)$;

$$1.000000000 \quad \textbf{(15)}$$

> $plot(H, \rho = 0..4 \cdot 10^{12})$;


1.5 × 10^23
1. × 10^23
5. × 10^22
0
-5. × 10^22
-1. × 10^23
-1.5 × 10^23
1. × 10^12
2. × 10^12
3. × 10^12
4. × 10^12
ρ